



Bumble in Space

The NASA-built robot, Bumble, flew on the International Space Station for the first time recently. <https://digit.in/july19-54>



NASA to study the sun

NASA declared that they will launch two missions to study the sun and its effects on space weather. <https://digit.in/july19-55>

Modern physics believes that time is very much real. However, the flow of time is what is the illusion. The flow of time is based on our perceptions, and seems to always flow from the past to the future. Physicists believe that we live in only one moment, call it 'now', and because we have memories of earlier moments, we assume that time changes, but maybe it doesn't. It's just us who are moving through spacetime.

While you're watching a movie for the first time, it feels like the present. The movie was probably shot months or even years ago, it still feels like the present when you are watching it for the first time. Now, if you watch the movie again immediately, it will always remind you of the past, or the first time you watched the movie.

Through all of that, outside of your own little conscious bubble, what really changed? If someone else came in to your little universe and looked at you and your phone, they would find no difference in file size, and no real difference in you. Yet your mind perceives a passage of time, a change in knowledge, an experience that is essentially invisible outside of your closed room. It's why we have parents and significant others bugging us for 'doing nothing the whole day' and yet we feel like we have done a lot... The next time that happens, shrug your shoulders as you exclaim, "Crazy huh? Time is relative, and maybe an illusion. So let's not sweat the small stuff..." (Remember to duck when stuff is thrown at you).

QUALIA

There's a concept called qualia that brings absolutely everything into question. This is more of a philosophical mystery than a scientific one, but we couldn't resist adding it here. American philosopher, C.I. Lewis, first used it in the modern way in his book 'Mind and the World Order'.

Think of qualia as the experiences that you have with the world. Now, in order to live in society, we have to agree on the validity of experiences. You are sitting on a bench and I walk

up and punch you in the nose, there is a shared experience not just between us, but also with all the bystanders. When the police arrive, and you and other bystanders describe what I did, I'm going to jail. This is because it is a shared experience, and everyone's qualia match. However, how do we know that everything does match?

Take for example two people (A and B) who have only one major difference – while one sees colours normally, the other sees them inverted. Both have grown up learning the same name for colours, and thus A thinks the background of our logo is "red" and B thinks of the inverse (what appears light blue) as "red" also. The same applies for dark, light, and everything else. Since there's no disagreement between the two ever, and both always see each other use the correct names for colours, they would think that the world looks identical to both of them. Yet, we know that if A was shown what B sees, he would certainly disagree with B. This is the simplest way we could think of to illustrate the qualia problem, which essentially calls into question reality itself.

MISSING LITHIUM

The Big Bang Theory is widely accepted as the best explanation we have for the origin story of our universe. However, the BBT does, in fact, have large holes that still need to be filled before we can consider it to be as accurate a theory as, say, evolution is.

One of those rather large holes in the theory is the problem of Lithium. According to the theory, for up to 20 minutes after the Big Bang occurred, the universe was expanding rapidly, and was very, very hot. Atomic nuclei were starting to form at this time, but because of the immense heat, were unable to form bonds with electrons to actually form atoms. According to the theory, only nuclei of lighter elements such as Helium, Deuterium and Lithium should have formed, as it would need fusion to form heavier elements, which would occur much later when stars formed and eventually died.

Based on observations, the BBT theory is spot on for Hydrogen and Helium, but is off by a mile when it comes to the amount of Lithium observed in the universe. Observations suggest that there is 3 or 4 times less lithium in the universe than we expect to see. However, many studies are proposing explanations for why we see so little lithium, and though none have been elevated to the level of a theory yet, it's hoped that one of them might be based on observational data.

WHAT'S INSIDE A BLACK HOLE?

A black hole is essentially an object with such a high mass in such a small volume, that it pretty much defies the laws of physics as we know them. So strong is its gravitational attraction that even light cannot escape it. Of course, since gravity reduces with the square of the distance to the object it is acting on, there is a point in space from the 'singularity' where gravity is low enough for light to escape – any closer and even light gets sucked in. This distance from the singularity in space is called the event horizon of the black hole. Because all of the signals we are capable of detecting are physical, we cannot measure anything past the event horizon of a black hole.

Why would we want to? Because beyond the event horizon of a black hole might lie the clue to laws and new physics that we haven't even thought of. We'd need to be able to 'look into' a black hole's event horizon to understand the very beginning of our universe. Inside this event horizon, or even inside the singularity itself might lie the answers to multiple questions, such as what happened before the big bang, and was there a 'before' to even consider. We might be able to answer questions about multiple universes, or whether there is a point at which the forces we observe in nature can be combined into one unifying force – this would lead to a unified theory of everything, and finally bring a relation between the other forces and gravity.

The problem is, the nearest black holes are too far away for even probes